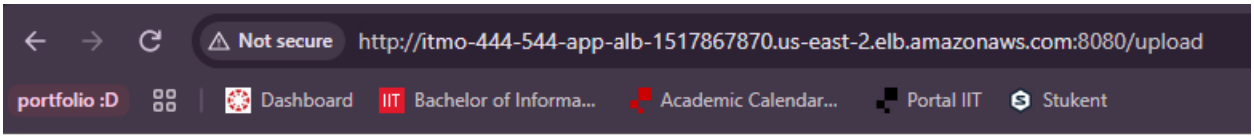
1.

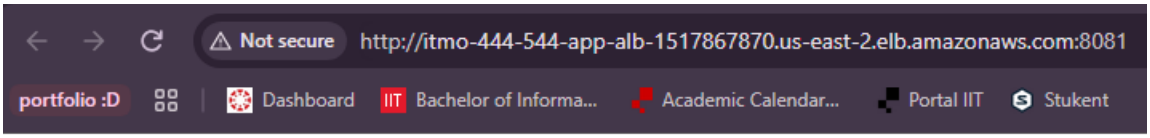
```
Done. ALB DNS name: itmo-444-544-app-alb-1517867870.us-east-2.elb.amazonaws.com
Uploader: http://itmo-444-544-app-alb-1517867870.us-east-2.elb.amazonaws.com:8080/
Viewer: http://itmo-444-544-app-alb-1517867870.us-east-2.elb.amazonaws.com:8081/
Give the target groups 1-2 minutes to report healthy - check with:
 aws elbv2 describe-target-health --target-group-arn arn:aws:elasticloadbalancing:us-east-2:452857281055:targetgroup/itm
o-444-544-uploader-tg/c5803b197af84f07 --region us-east-2
Getting instance IDs from Auto Scaling Groups...
Waiting for i-036300b9aa715d820 to reach running...
-----------------------------------------------------
|               DescribeInstances                   |
+----------------------+----------+-----------------+
|  i-036300b9aa715d820 |  running |  3.15.183.83    |
+----------------------+----------+-----------------+
Waiting for i-08b582694bb0088cc to reach running...
-----------------------------------------------------
|               DescribeInstances                   |
+----------------------+----------+-----------------+
|  i-08b582694bb0088cc |  running |  3.144.102.164  |
+----------------------+----------+-----------------+
```



Upload successful.

Upload another

2.



Viewer

Uploaded file: shared.txt

```
thank you for trusting me professor :)
```

Refresh

3.

```
root@334708184 0f6:/lab04# aws elbv2 describe-target-health \
  --target-group-arn arn:aws:elasticloadbalancing:us-east-2:452857281055:targetgroup/itmo-444-544-uploader-tg/c5803b197a
f84f07 \
  --region us-east-2 \
  --query 'TargetHealthDescriptions[*].[Target.Id,TargetHealth.State]' \
  --output table

aws elbv2 describe-target-health \
  --target-group-arn arn:aws:elasticloadbalancing:us-east-2:452857281055:targetgroup/itmo-444-544-viewer-tg/9b76d4b1cd35
6b6c \
  --region us-east-2 \
  --query 'TargetHealthDescriptions[*].[Target.Id,TargetHealth.State]' \
  --output table
-----------------------------------
|       DescribeTargetHealth      |
+----------------------+----------+
|  i-036300b9aa715d820 |  healthy |
+----------------------+----------+
-----------------------------------
|       DescribeTargetHealth      |
+----------------------+----------+
|  i-08b582694bb0088cc |  healthy |
+----------------------+----------+
```

4.
```
root@3347081840f6:/lab04/testing# python3 chaos_test.py --tier uploader
Target: uploader tier at http://itmo-444-544-app-alb-1517867870.us-east-2.elb.amazonaws.com:8080/
Confirming the endpoint is healthy before we start...
Terminating i-036300b9aa715d820 via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_uploader_802.csv
Recovery time: 83.8s (SLO: 180s)
PASS: recovery time met the SLO.
root@3347081840f6:/lab04/testing#
```
5.
```
root@3347081840f6:/lab04/testing# python3 chaos_test.py --tier uploader
Target: uploader tier at http://itmo-444-544-app-alb-1517867870.us-east-2.elb.amazonaws.com:8080/
Confirming the endpoint is healthy before we start...
Terminating i-0b0e8bb9296a7e7d9 via the Auto Scaling Group...
Waiting for the endpoint to go down...
Waiting for the endpoint to come back up...
Poll log written to chaos_test_uploader_3537.csv
FAIL: endpoint did not recover within 600s.
```
6. The recovery window was mainly affected by the time it took for the EC2 instances to launch, install dependencies, and start the application, rather than just the load balancer health checks. If booting and installing dependencies took about 120 seconds and health checks added another 30 seconds, the total recovery time would be approximately 150 seconds. This means boot and application startup dominated the recovery window, while health-check timing added some extra delay before the application was considered healthy.
7. In create_app_stack and delete_app_stack, the scrips were updated to support Auto Scaling Groups instead of relying on the previous lab's direct run-instances command. The direct instance-launch command was replaced with the creation and configuration of launch templates and Auto Scaling Groups. This allows AWS to manage the EC2 instances, replace unhealthy instances, and maintain the desired number of instances. The deletion script was updated to remove the Auto Scaling Groups before deleting the load balancer, target groups, and related security resources.
8.
```
== Deleting Auto Scaling Groups (this also terminates their instances) ==
Waiting for both Auto Scaling Groups to be gone...
itmo-444-544-uploader-asg deleted.
itmo-444-544-viewer-asg deleted.
== Deleting the load balancer ==
== Deleting target groups ==
== Restoring the app security group to its previous-lab state ==
== Deleting the ALB security group ==
== Removing IAM instance profiles and roles ==
== Emptying and deleting S3 bucket ==
delete: s3://itmo-444-544-vabill-fileshare/shared.txt
Cleanup complete.
```